

response to pathogens in body fluids versus in body cells, and emphasizes the central role of the helper T cell.

Immunization

The protection provided by a second immune response provides the basis for **immunization**, the use of antigens artificially introduced into the body to generate an adaptive immune response and memory cell formation. In 1796, Edward Jenner noted that milkmaids who had cowpox, a mild disease usually seen only in cows, did not contract smallpox, a far more dangerous disease. In the first documented immunization (or *vaccination*, from the Latin *vacca*, cow), Jenner used the cowpox virus to induce adaptive immunity against the closely related smallpox virus. The *vaccines* used today for immunizations may be made from inactivated bacterial toxins, killed or weakened pathogens, or even genes encoding microbial proteins. Because all of these agents induce a primary immune response and immunological memory, an encounter with the pathogen from which the vaccine was derived triggers a rapid and strong secondary immune response (see Figure 43.17).

Vaccination programs have been successful against many infectious diseases that once killed or incapacitated large numbers of people. A worldwide vaccination campaign led to eradication of smallpox in the late 1970s. In industrialized nations, routine immunization of infants and children has dramatically reduced the incidence of sometimes devastating diseases, such as polio and measles (**Figure 43.24**). Unfortunately, not all pathogens are easily managed by vaccination. Furthermore, some vaccines are not readily available in impoverished areas of the globe.

Misinformation about vaccine safety and disease risk has led to a growing public health problem. Consider measles as just one example. The vaccine is highly effective and very safe—fewer than one in a million children experience a significant allergic reaction. Indeed, a vaccination campaign had effectively eliminated the disease in the United States by 2000. However, the disease remains quite dangerous to this day, killing more than 200,000 people worldwide each year. Recently, a decline in vaccination rates in the United States has led to multiple outbreaks since 2010. A multi-state outbreak in 2019 was the worst in two decades, with more than 1,100 cases in the first six months of

the year. Even with modern medical care, measles kills on average one out of every 1,000 infected individuals.

Active and Passive Immunity

Our discussion of adaptive immunity has to this point focused on **active immunity**, the defenses that arise when a pathogen infection or immunization prompts an immune response. A different type of immunity results when, for example, the IgG antibodies in the blood of a pregnant female cross the placenta to her fetus. This protection is called **passive immunity** because the antibodies in the recipient (in this case, the fetus) are produced by another individual (the mother). IgA antibodies present in breast milk provide additional passive immunity to the infant's digestive tract while the infant's immune system develops. Passive immunity persists only as long as the transferred antibodies last (up to a few months).

In artificial passive immunization, antibodies from an immune animal are injected into a nonimmune animal. For example, humans bitten by venomous snakes are sometimes treated with antivenin, serum from sheep or horses that have been immunized against a snake venom. When injected immediately after a snakebite, the antibodies in antivenin can neutralize toxins in the venom before the toxins do massive damage.

▼ **Figure 43.24 Vaccine-based protection against two life-threatening communicable diseases.** The graphs show deaths by year in the United States caused by polio and measles. The maps show examples of the global progress against these two diseases.

